

Learning-Based Scheduling and Motion Coordination of Mobile Robots in Smart Factories

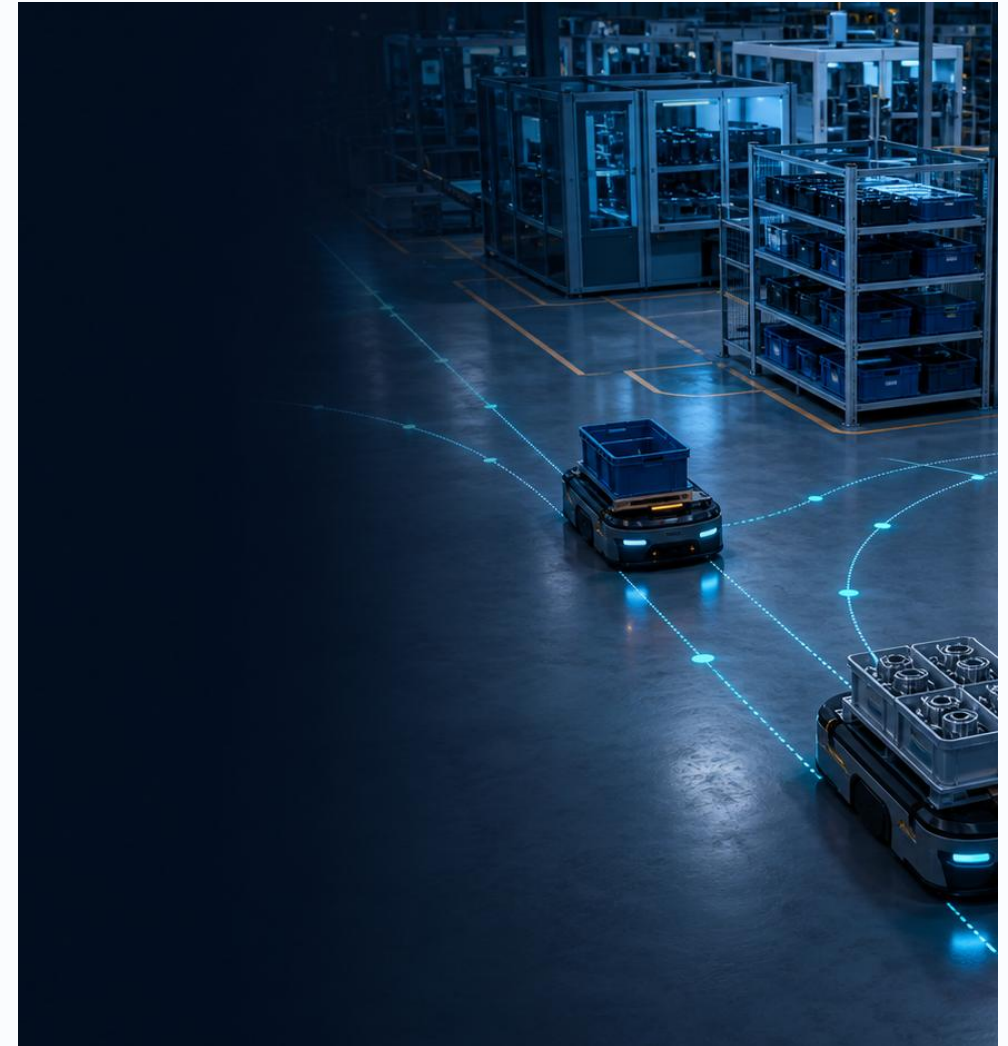
A simulation-based study of task allocation, coordinated routing and safe multi-robot execution

Wenbin Lin | H00484764
MSc Robotics
Heriot-Watt University Dubai



- Flexible factories replace fixed conveyors with mobile robot fleets.
- Dynamic transport requests compete for robots, corridors, docks and charging resources.
- A locally efficient assignment may create global congestion or unsafe encounters.
- The research challenge is to improve throughput without separating scheduling from physical execution.

Core idea · allocate intelligently, route safely, execute synchronously



How can learning-based scheduling improve factory productivity and robot coordination compared with rule-based approaches?

01

Model

Build a reproducible smart-factory simulator with dynamic transport demand.

02

Schedule

Implement rule-based, optimisation, metaheuristic and reinforcement-learning schedulers.

03

Coordinate

Integrate assignment with grid routing, reservations, local avoidance and recovery.

04

Evaluate

Compare throughput, waiting time, decision latency, safety and workload balance.

Scope decision: Webots is the implemented validation platform; ROS 2 / Gazebo is positioned as a future portability and sim-to-real extension.

A Small fleet



3 robots

Mean arrival: 30 s

≈ 2 tasks/min

Poisson arrivals

B Medium fleet



5 robots

Mean arrival: 15 s

≈ 4 tasks/min

Poisson arrivals

C Large fleet



8 robots

Mean arrival: 8 s

≈ 7.5 tasks/min

Poisson arrivals



R Robot platform

Differential drive · 0.26 m/s max · wheel radius 0.033 m · wheelbase 0.287 m

T Task types

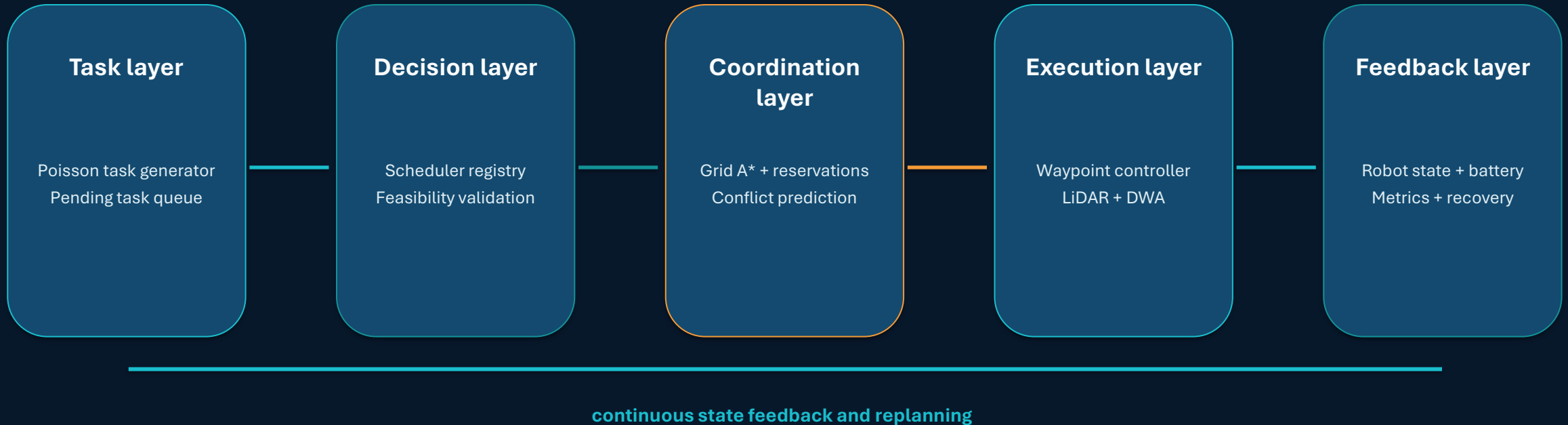
Storage → workstation · workstation → storage · workstation → workstation

E Energy model

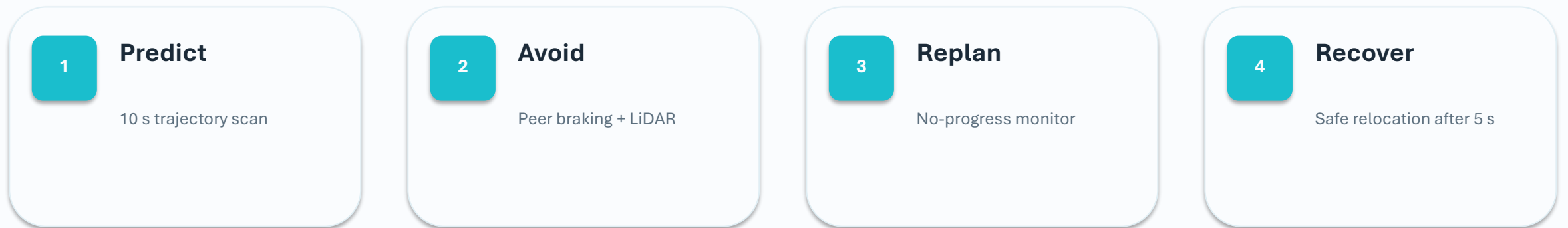
25% low-battery trigger · 15% task-abort threshold · charge to 95%

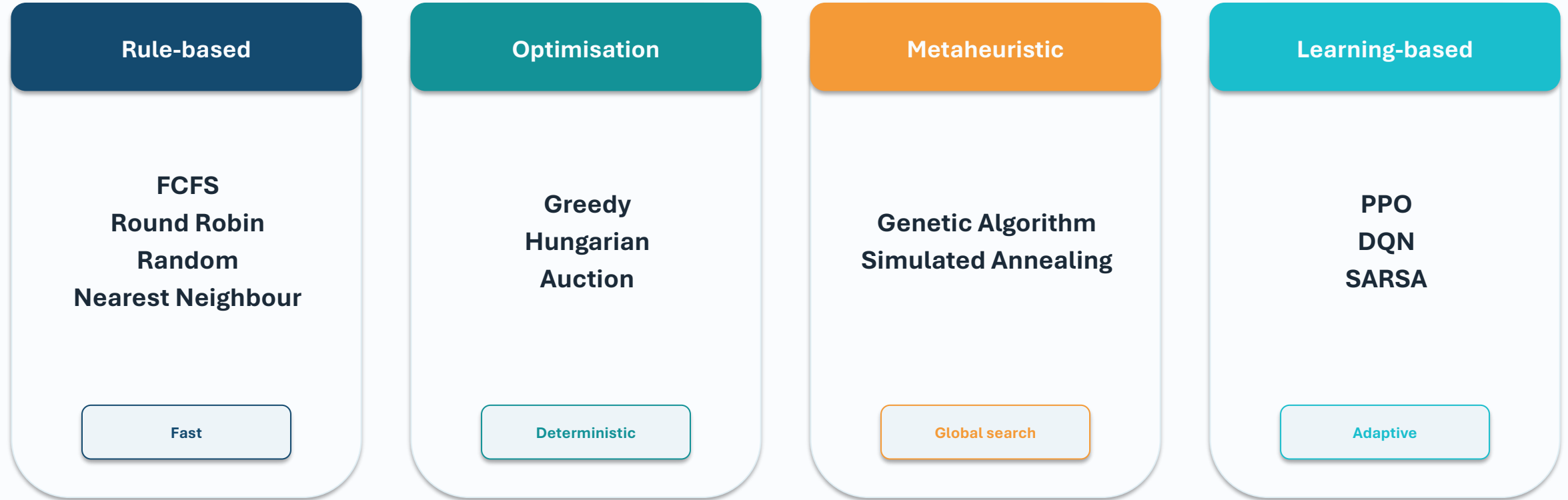
D Demand model

Valid pickup–delivery pairs generated by an exponential inter-arrival process



Design principle: a scheduling decision is committed only after a feasible path exists and the navigation command is successfully dispatched.





Current default usage

Common runtime choice: FCFS / Hungarian-style safe fallback · Research comparison: GA, SA, Auction and RL policies

Five reinforcement-learning options

HOW LEARNING CAN SUPPORT ROBOT SCHEDULING

PPO

Stable policy-gradient training

Large continuous state representations

DQN

Value learning with replay

Discrete robot–task assignment actions

SARSA

On-policy temporal-difference

Risk-aware learning under the behaviour policy

A2C

Actor–critic with low variance

Fast online policy/value updates

QMIX / MAPPO

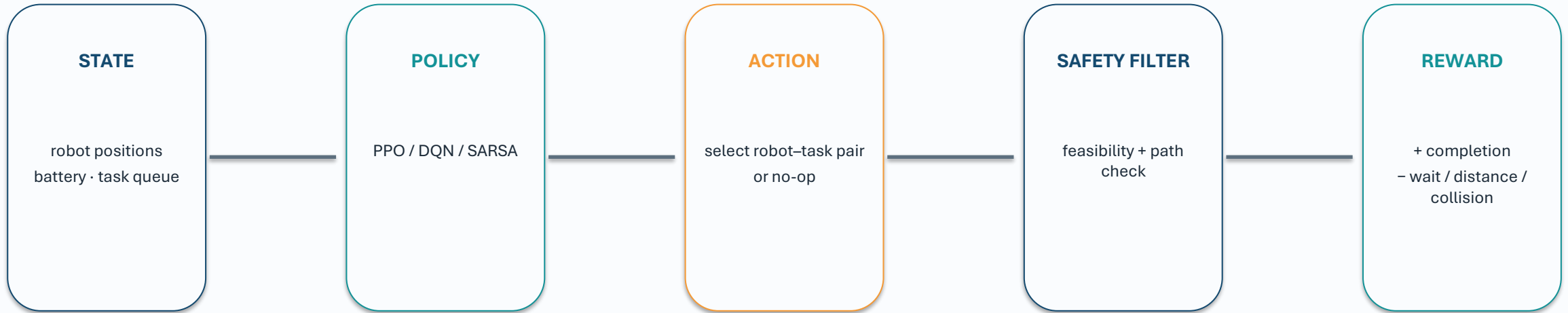
Cooperative multi-agent learning

Decentralised robots with a shared objective

Implemented in this project: PPO, DQN and SARSA · A2C and cooperative MARL are proposed extensions

Reinforcement-learning loop

STATE → ACTION → SAFE EXECUTION → REWARD



next state and long-term return

Why learning helps

Optimises delayed effects: queue growth, future travel and resource contention.

Why classical control remains

Guarantees executable paths and reacts to centimetre-scale hazards.

Hybrid principle

Learning proposes; validation and motion coordination enforce safety.

Inputs held constant

Same factory layout
Same robot dynamics
Same task-generation model
Same coordination layer

Primary metrics

Throughput · completion rate
Waiting / completion time
P95 scheduling latency
Workload CV · safety distance

Experimental protocol

30-minute simulation horizon
Fixed scenario definitions
Recorded JSON outputs
Multi-seed repetition planned

CURRENT EVIDENCE

Scenario A contains nine baseline runs plus one preliminary PPO_RL deployment run.

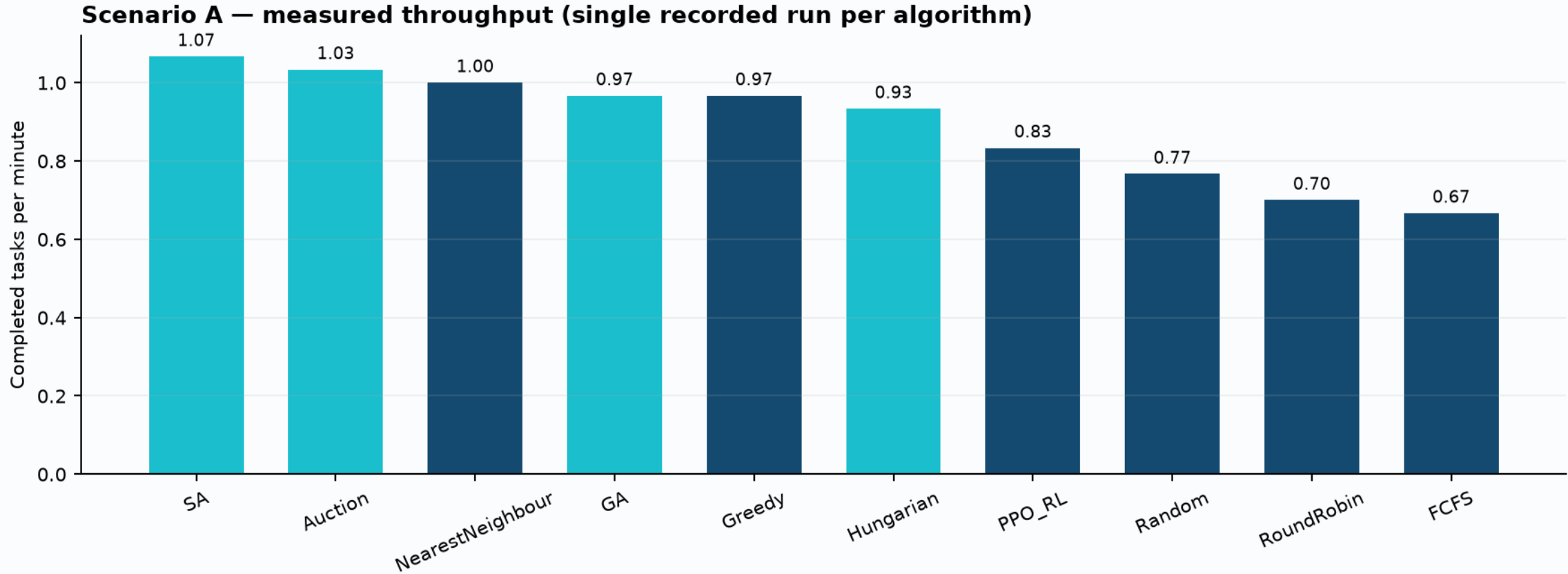
OPEN EVIDENCE GAP

Matched multi-seed RL evaluation across Webots A/B/C is not yet complete.

Therefore, measured claims and expected RL benefits are reported separately.

Measured scheduling performance

SCENARIO A — NINE BASELINES PLUS PPO_RL

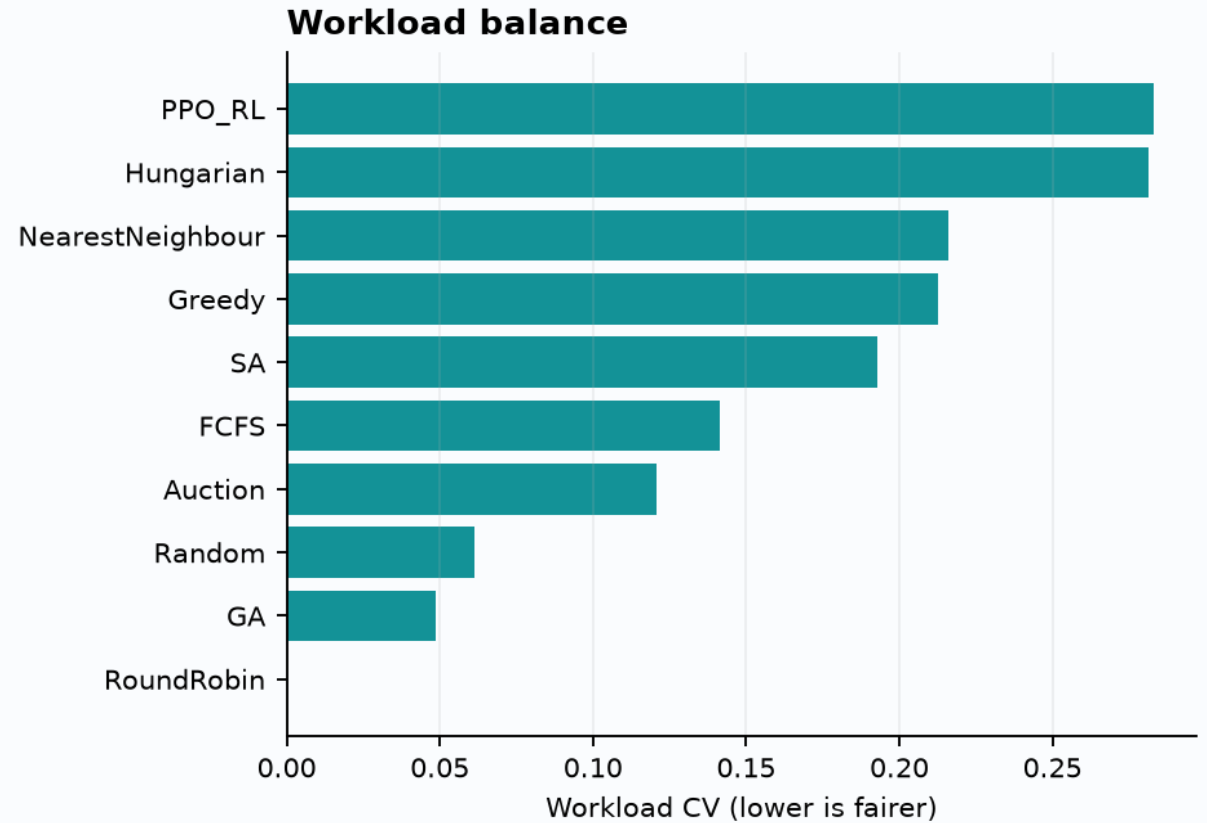
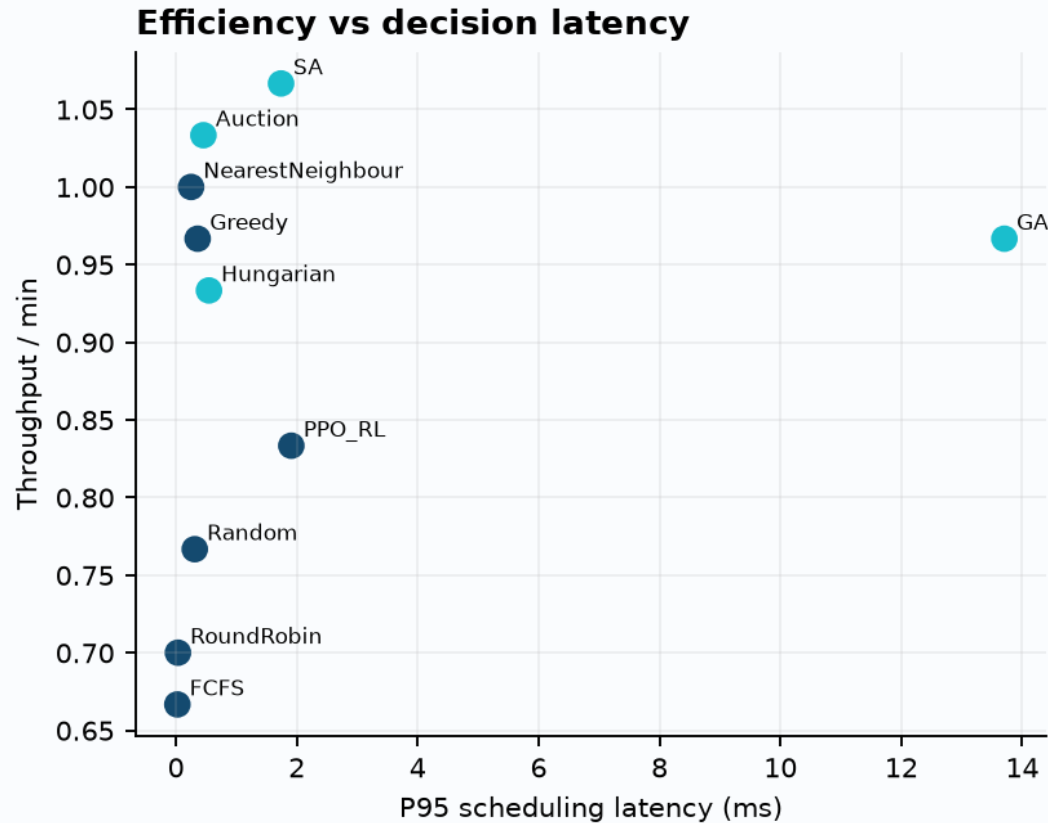


Highest recorded: SA · 1.07 tasks/min

Interpretation is preliminary: each bar is one recorded 30-minute run, not yet a balanced multi-seed confidence interval.

Efficiency is not a single number

LATENCY, THROUGHPUT, FAIRNESS AND SAFETY TRADE-OFFS



Observed pair-distance violations: 21

A production scheduler must deliver useful assignments within the controller budget and must not destabilise the shared motion layer.

14 Traditional scheduling vs reinforcement learning

MEASURED TRAINING EVIDENCE AND EXPECTED SYSTEM-LEVEL ADVANTAGES

Learning curves in the abstract scheduling environment



Potential RL advantage

- Optimises long-term return rather than one-step distance.
- Can encode battery, queue pressure and future congestion.
- Inference is fast after offline training.
- Adapts policy behaviour across changing demand patterns.

Evidence boundary

Claims require matched multi-seed Webots A/B/C tests against the baselines.

Research position: RL should improve assignment quality; classical coordination remains the safety envelope.

A hybrid learning-and-control framework

- The project integrates dynamic scheduling with executable multi-robot motion coordination.
- Traditional algorithms provide interpretable, measurable baselines and robust fallbacks.
- PPO, DQN and SARSA establish the learning pipeline; matched Webots comparison is the next milestone.
- Future work: repeated A/B/C trials, confidence intervals, curriculum learning, MARL, ROS 2 / Gazebo portability and hardware validation.

Thank you · Questions and discussion

